

Laplace Transform Notes

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1. What's the Point?

Differential equations are annoying to solve. Piecewise/discontinuous functions are also annoying. The Laplace transform is here to save the day!

2. Definition

Definitions

The *Laplace transform* is an operation that takes a function of a real variable (usually t) as input and outputs a function of a complex variable s . If $f(t)$ is our original function, $F(s)$ is its transform.

$$\mathcal{L}\{f(t)\}(s) = F(s) \stackrel{\text{def}}{=} \int_0^{\infty} f(t)e^{-st} dt$$

There is a slight caveat. The above integral may not converge. We will ignore this for now.

Example 1

Calculate $\mathcal{L}\{e^{at}\}$.

$$\begin{aligned}\mathcal{L}\{e^{at}\} &= \int_0^{\infty} e^{at}e^{-st} dt \\ &= \frac{1}{s-a}\end{aligned}$$

This is not exactly correct. The integral converges only when $\Re(s) > a$. But we will wave our hands and simply say

$$\mathcal{L}\{e^{at}\}(s) = \frac{1}{s-a} \text{ for } s \neq a$$

3. Properties of the Laplace Transform

3.1. Linearity

Linearity

The Laplace transform is linear!

$$\mathcal{L}\{af + bg\} = a\mathcal{L}\{f\} + b\mathcal{L}\{g\}$$

which follows from the linearity of integrals.

3.2. Existence

Existence Theorem

A function f is of *exponential order* if $|f(t)| \leq Me^{ct}$ for some positive M and c .

If f is continuous and of exponential order with constant c , then $\mathcal{L}\{f\}$ exists for all $s > c$.

Proof follows by comparison.

3.3. Unique and Invertible

Existence Theorem

Two integrable functions have the same Laplace transform if they differ on a set of Lebesgue measure zero. For example, if two functions are different at only finitely many points, they have the same Laplace transform.

We will define an inverse operation

$$\mathcal{L}^{-1}\{F(s)\} = f(t)$$

4. Some Useful Laplace Transforms

Table of transforms

Assume f is sufficiently nice.

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a} \text{ for } s \neq a$$

$$\mathcal{L}\{\sin at\} = \frac{a}{s^2 + a^2}$$

$$\mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2}$$

$$\mathcal{L}\{t^n\} = \frac{\Gamma(n+1)}{s^{n+1}}$$

$$\mathcal{L}\{\ln t\} = \frac{\ln s + \gamma}{s}$$

$$\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s} \text{ where } u \text{ is the Heaviside step function}$$

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}$$

$$\mathcal{L}\{u(t-a)f(t-a)\} = e^{-as}F(s)$$

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-sp}} \int_0^p e^{-s\tau} f(\tau) d\tau \text{ where } p \text{ is the period}$$

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - \dots - sf^{(n-2)}(0) - f^{(n-1)}(0)$$

$$\mathcal{L}\left\{\int_0^\tau f(\tau) d\tau\right\} = \frac{F(s)}{s}$$

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a) \text{ (the shift property)}$$

$$\mathcal{L}\{f * g\} = F(s)G(s)$$

$$\mathcal{L}\{tf(t)\} = -F'(s)$$

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s') ds'$$

5. Misc Notes

$\mathcal{L}\{f'\} \rightarrow$ use by parts

$\mathcal{L}\left\{\int_0^t\right\} \rightarrow$ change order of integration, otherwise convolution

$$(f * g)(t) \stackrel{\text{def}}{=} \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau$$

$$\mathcal{L}\{f * g\} = F(s)G(s)$$

Set $G = 1$ (the step function) and the result follows.

This holds for nested integrals!

$$\mathcal{L}\left\{\int^n\right\} = \frac{F(s)}{s^n}$$

This is true by Cauchy's repeated integration formula, and then convolution theorem.

Feynman's trick!

$$\mathcal{L}\{tf(t)\} = -F'(s)$$

This generalizes into $t^n f(t)$.

We want to prove $\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s')ds'$. We will use the same Feynman's trick, again!

$$\begin{aligned} Q(s') &= \int_0^\infty e^{-st} \frac{f(t)}{t} dt \\ \frac{dQ}{ds'} &= - \int_0^\infty e^{-st} f(t) dt = F(s) \\ \int dQ &= - \int F(s') ds' \end{aligned}$$

Now we want to set appropriate bounds for the above integral. One of the bounds should be s , the variable we want. What should the other one be? 0 doesn't work, the next thing we try is ∞ , since that makes the exponential term zero.

$$\begin{aligned} \int_s^\infty dQ &= - \int_s^\infty F(s') ds' \\ Q(s) &= \int_s^\infty F(s') ds' \end{aligned}$$

Now let's try $\mathcal{L}\{\ln t\}$.

$$\begin{aligned} \int_0^\infty e^{-st} \ln t dt &= \lim_{n \rightarrow \infty} \int_0^{\frac{n}{s}} \left(1 - \frac{st}{n}\right)^n \ln t dt \\ &= \lim_{n \rightarrow \infty} -\frac{n}{s} \int_0^1 x^n \ln\left(\frac{n(1-x)}{s}\right) dx \\ &= \lim_{n \rightarrow \infty} -\frac{n}{s} \left(\int_0^1 x^n \ln(1-x) dx + \int_0^1 x^n \ln\left(\frac{n}{s}\right) dx \right) \\ &= \lim_{n \rightarrow \infty} -\frac{n}{s} \left(- \int_0^1 x^n \sum_{j=1}^\infty \frac{x^j}{j} dx + \ln\left(\frac{n}{s}\right) \frac{1}{n+1} \right) \\ &= \lim_{n \rightarrow \infty} -\frac{n}{s} \left(- \sum_{j=1}^\infty \int_0^1 \frac{x^{j+n}}{j} dx + \ln\left(\frac{n}{s}\right) \frac{1}{n+1} \right) \\ &= \lim_{n \rightarrow \infty} -\frac{n}{s} \left(- \sum_{j=1}^\infty \frac{1}{j(j+n+1)} + \ln\left(\frac{n}{s}\right) \frac{1}{n+1} \right) \end{aligned}$$

Now let us look at that infinite sum

$$\begin{aligned}
\sum_{j=1}^{\infty} \frac{1}{j(j+n+1)} &= \frac{1}{n+1} \sum_{j=1}^{\infty} \left(\frac{1}{j} - \frac{1}{j+n+1} \right) \\
&= \frac{1}{n+1} \sum_{j=1}^{n+1} \frac{1}{j} \\
&= \frac{H_{n+1}}{n+1}
\end{aligned}$$

For large n , $H_n \sim \ln(n) + \gamma$. Using this,

$$\begin{aligned}
\mathcal{L}\{\ln t\} &= \lim_{n \rightarrow \infty} -\frac{n}{s} \left(-\sum_{j=1}^{\infty} \frac{1}{j(j+n+1)} + \ln\left(\frac{n}{s}\right) \frac{1}{n+1} \right) \\
&= \lim_{n \rightarrow \infty} \frac{n \ln\left(\frac{n}{s}\right) - \ln(n) - \gamma}{n+1} \\
&= \boxed{\frac{\ln s + \gamma}{s}}
\end{aligned}$$

Alternate method:

$$\begin{aligned}
\mathcal{L}\{t^n\} &= \frac{\Gamma(n+1)}{s^{n+1}} \\
\frac{d}{dn} \int_0^{\infty} e^{-st} t^n dt &= \frac{d}{dn} \frac{\Gamma(n+1)}{s^{n+1}} \\
\frac{d}{dn} \int_0^{\infty} e^{-st} t^n \ln t dt &= \frac{s^{n+1} \Gamma'(n+1) - \Gamma(n+1) s^{n+1} \ln s}{s^{2n+2}}
\end{aligned}$$

Set $n = 0$ and compute to get the result. However $\Gamma'(1)$ is a little tricky to get, and needs the same series expansion as above.